



Maritime &
Coastguard
Agency

Guidance

Annex 2 - Competence Standards for Ship Assist Towage Endorsement

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A2.1 Summary

This standard is about:

The competence required to conduct ship assist towage operations (assisting with the berthing and un-berthing of vessels) at a management level on board vessels operating in categorised waters or in limited coastal areas.

The standards for the **Ship Assist Towage Endorsement** are in addition to the requirements of the **General Towage Endorsement**.

This standard is for:

Masters of vessels engaged in ship assist towage operations in categorised waters or in limited coastal areas.

A2.2 Outcomes of effective performance

Candidates for the issue of the Ship Assist Towage Endorsement will have demonstrated that they can do the following:

1. Ensure that there are clearly defined safe work procedures/standing orders for conducting safe ship assist towage operations.
2. Assess the vessel to be provided assistance and establish correctly that it is safe to provide assistance to the planned destination.
3. Inform the relevant authority promptly where you assess the vessel to be unfit for receiving assistance, stating the reasons clearly.
4. Where necessary, ensure that all statutory, regulatory and other requirements for ship assist towage operations are identified and met, and comply with all navigational, reporting and warning requirements.
5. Assess the circumstances affecting the requirement for providing assistance and determine correctly any risks involved.
6. In conjunction with the pilot and/or vessel master, as appropriate, plan the ship assist operations, identifying correctly the actions required, taking into account the characteristics of the vessel to be assisted, and any relevant navigational restrictions, determining the best method for providing the assistance.

7. Establish and maintain clear and appropriate communications between your vessel and the vessel to which the assistance is being provided.
8. Brief all concerned clearly and correctly on the actions to be taken, ensuring that everyone understands their role and the emphasis upon safety.
9. Prepare correctly all necessary equipment for providing ship assist services, checking for damage prior to commencing operations, making good and reporting any faults before use
10. Develop and action appropriate contingency plans that ensure the safety of navigation, protection of the marine environment and safety of the vessel and persons on board during ship assist operations.
11. Ensure that the correct audible and visual signals indicate ship assist operations.
12. Handle the vessel safely under the prevailing conditions with due consideration to the engineering systems.
13. Operate remote controls of propulsion plant, engineering systems and services correctly and safely.
14. Monitor the tow and make any necessary adjustments correctly and safely.
15. Establish communications with those receiving the tow including the pilot and/or vessel master and other relevant authorities and ensure that preparations for receiving the tow on arrival are in place.
16. React safely and appropriately to loss of critical systems e.g. main power or steering unit whilst in close proximity/made fast to assisted vessel.
17. Disengage the tow correctly and safely.

A2.3 Knowledge and understanding

Underpinning Knowledge Syllabus for Ship Assist Towage Endorsement

Note: Whilst some of the knowledge areas below are similar to the General Towage Endorsement, they are intended to reflect a greater depth of knowledge in these specific areas.

Candidates need to know and practically demonstrate the following:

SAT 1 Ship Assist, Tug Handling Knowledge

1.1 The limitations of point load on ships' hulls and how this might affect a ship assist operation.

1.2 The functions and limitations of different fendering arrangements.

1.3 The dangers and pressure areas arising from the construction of ships when operating in close proximity e.g. interaction forces, lines of sight.

1.4 The correlation between the windage of a vessel, the expected weather conditions and the bollard pull of the tug.

1.5 How to react to loss of critical systems whilst in close proximity/made fast to assisted vessel.

1.6 The safe operation of the vessel in port in restricted visibility whilst conducting ship assist operations.

1.7 A safe departure from berth and on completion, conduct a berthing alongside.

1.8 The ability to manoeuvre effectively under free sailing conditions.

1.9 Secure a barge alongside from a mooring and get underway.

1.10 How to manoeuvre and re-secure a lighter/barge.

1.11 Take station as the bow tug on large ship movement, including a running catch up from the shoulder.

1.12 Act as stern tug on large ship movement, including a running catch up.

1.13 To secure as the alongside tug, including changing sides within a winding/re-berth manoeuvre.

1.14 The role of the push/pull tug in a berthing or sailing movement.

1.15 How to act as lead bow tug into and/or out of tidal basin.

1.16 How to pick up bow to bow (ASD) 1.17 Typical abort manoeuvres and recovery position

1.18 Active winch control

1.19 The hazards of using towed vessel's ropes, gear and securing points.

SAT 2 External Communications

2.1 The importance of the need for a clear berthing plan and agreement by all parties.

2.2 Communications between tug(s) and pilot:

- Use standard communications protocols / instructions for power and direction.
- Appreciate the importance of locally agreed terminology and means of communications.
- Clearly identify and establish communications with multiple tugs.

SAT 3 Emergency Procedures

3.1 Reacting to loss of critical systems e.g. main power or steering unit whilst in close proximity/made fast to assisted vessel.

3.2 Reacting to malfunction of critical systems on assisted vessel.

3.3 Methods of slipping tows under load and the associated hazards.

SAT 4 Fitness for Purpose

4.1 Understanding the factors that influence the correct choice of tug (bollard pull, propulsion type, manoeuvrability, size of vessel) for a given towage operation.

4.2 Appropriate use of available tugs.

4.3 Ensuring tug watertight integrity during towage operations.

SAT 5 Detailed Local Knowledge

Detailed knowledge of:

5.1 Harbour and terminal layout, to include:

- Berths, navigational channels and buoyage
- Local tidal flows and currents
- Limiting depths in berthing areas

5.2 Port Regulations including local pilotage requirements, local byelaws, towing in restricted visibility and towage guidelines.

5.3 Relevant parts of the Port Marine Safety Code relating to towage.

5.4 Vessel traffic Services, traffic situational awareness and port movements.

SAT 6 Safe Towage Operation

6.1 Tug Handling

Thorough knowledge of tug handling in a range of Ship Assist manoeuvres including:

- a. Direct and Indirect towing
- b. Push and Pull techniques
- c. Picking up and letting go over the bow or stern
- d. Escort towing – steering and braking techniques
- e. Securing alongside and manoeuvring dead ship vessels.

6.2 Interaction

A detailed understanding of interaction with particular reference to working in close proximity with large vessels, especially in bow to bow work with Azimuth Stern Drive (ASD) and Azimuth Tractor Drive (ATD) tugs.

Reference: CaptHenkHensen article on Use of ASD Tugs - ISBN 1 870077 73 3 published by the Nautical Institute – 2006

6.3 Fendering

- a. Understanding the effects and limitations of point load on ships' hulls.
- b. Understand the effects, functions and limitations of different fendering arrangements on tugs.

6.4 Stability and watertight integrity

- a. Appreciation of stability and the need for watertight integrity when towing under load conditions.
- b. Knowledge of towing points and the avoidance of girting in dynamic situations.

SAT 7 Towing Equipment

Practical knowledge of the capabilities and limitations of different types of the following as used in Ship Assist operations to include:

7.1 Winches and Gear

- a. Towing winches and capstans - rendering and tensioning
- b. Towing hooks
- c. Tow ropes - wires, synthetic ropes – polyester, High Modulus PolyEthylene (HMPE) ropes etc.
- d. Use of pennants
- e. Use of grommets
- f. Use of joining shackles and connections

7.2 Use of assisted ship's ropes and gear.

- a. Safe working load and limitations of securing points
- b. Hazards and limitations in the use of ship's ropes

7.3 Testing and Inspection of towage equipment

- a. Quick release gear
- b. Records of certification of ropes wires and shackles
- c. Inspection and renewal procedures

SAT 8 Propulsion Systems and Tug Types

8.1 Detailed knowledge of capabilities and limitations (including power rating) of tug types to include:

- a. Voith Schneider
- b. Azimuth Stern Drive (ASD)
- c. Azimuth Tractor Drive (ATD)
- d. Conventional single screw e. Conventional twin screw

8.2 Outline knowledge of other tug types including:

- a. 'Z' Tech
- b. Rotor

c. Carousel Page



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